

This database design follows a relational approach to ensure data integrity, security, and scalability for the AI Fitness Coach platform.

1. Database Overview

The schema is designed to manage user authentication, biometric profiles, equipment availability, AI-generated plans, and progress tracking. It adheres to 3rd Normal Form (3NF) to minimize redundancy and maximize update performance.

2. ER Diagram Description

- **Users:** Core entity containing profile and auth data.
- **Equipment/User_Equipment:** Many-to-many relationship mapping available gym gear to users.
- **Plans/Workouts/Exercises:** Hierarchical structure where a User has a Plan, containing daily Workouts, composed of specific Exercises.
- **Logs/Measurements:** Time-series tables capturing user progress (weights, sets, body stats).

3. SQL Schema

```
-- Users and Profile
CREATE TABLE Users (
    user_id UUID PRIMARY KEY,
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash TEXT NOT NULL,
    age INT CHECK (age >= 18),
    gender VARCHAR(20),
    height_cm DECIMAL(5,2),
    fitness_level VARCHAR(50), -- Beginner, Intermediate, Advanced
    goals TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE Equipment (
    equipment_id SERIAL PRIMARY KEY,
    name VARCHAR(100) UNIQUE NOT NULL
);

CREATE TABLE User_Equipment (
    user_id UUID REFERENCES Users(user_id),
    equipment_id INT REFERENCES Equipment(equipment_id),
    PRIMARY KEY (user_id, equipment_id)
);

-- Plans and Content
CREATE TABLE Plans (
```

```

        plan_id UUID PRIMARY KEY,
        user_id UUID REFERENCES Users(user_id),
        generated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
        metadata JSONB -- Stores AI prompt context or versioning
    );

CREATE TABLE Daily_Workouts (
    workout_id UUID PRIMARY KEY,
    plan_id UUID REFERENCES Plans(plan_id),
    scheduled_date DATE NOT NULL,
    is_completed BOOLEAN DEFAULT FALSE
);

CREATE TABLE Workout_Logs (
    log_id UUID PRIMARY KEY,
    workout_id UUID REFERENCES Daily_Workouts(workout_id),
    exercise_name VARCHAR(100),
    sets INT,
    reps INT,
    duration_minutes INT,
    completed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Progress Tracking
CREATE TABLE User_Measurements (
    measurement_id SERIAL PRIMARY KEY,
    user_id UUID REFERENCES Users(user_id),
    weight_kg DECIMAL(5,2),
    body_fat_percentage DECIMAL(4,2),
    recorded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT one_per_day UNIQUE (user_id, CAST(recorded_at AS DATE))
);

```

4. Normalization Notes

- ****1NF:**** All columns contain atomic values; tables have defined Primary Keys.
- ****2NF:**** All non-key attributes are fully functionally dependent on the Primary Key.
- ****3NF:**** No transitive dependencies. For example, measurement units and equipment details are mapped to avoid repeating string values, using IDs instead.
- ****Business Logic Constraint:**** The `one\_per\_day` constraint on `User\_Measurements` enforces the rule that biometric data updates are limited to once per 24 hours.

5. Indexes

- `CREATE INDEX idx\_user\_measurements\_date ON User\_Measurements(user\_id, recorded\_at DESC);` (Optimizes dashboard chart rendering).

- ``CREATE INDEX idx_workouts_date ON Daily_Workouts(scheduled_date);`` (Optimizes daily notification engine queries).
- ``CREATE INDEX idx_users_email ON Users(email);`` (Optimizes login/auth).

6. Security and Constraints

- **Encryption:** Application layer handles PII encryption before storage.
- **PII:** Database strictly stores only essential physiological data; metadata is stored in ``JSONB`` for flexibility without schema changes.
- **Integrity:** Foreign keys ensure that orphaned workout logs or plans cannot exist.
- **Consistency:** The ``CHECK (age >= 18)`` constraint enforces the business rule at the database level.

7. Relationship Summary

- **One-to-Many:** ``Users`` → ``Plans`` → ``Daily_Workouts`` → ``Workout_Logs``.
- **Many-to-Many:** ``Users`` ↔ ``Equipment`` (via ``User_Equipment`` junction table).
- **One-to-Many:** ``Users`` → ``User_Measurements``.